

APPENDIX A

ILLUSTRATIVE EXAMPLE FOR ALGORITHM 1

For clarity, we use an example with $|\mathcal{I}| = 4$ clients and $K = 2$ to describe Algorithm 1, as shown in Figure 5. Suppose the set of selected clients is $\{1, 2\}$ at time slot t . The selected clients collect fresh data $D_{i,t}, \forall i \in \{1, 2\}$ such that $D_{i,t} \cap D_{i,t-1} = \emptyset$, while the remaining clients $\{3, 4\}$ keep the same dataset $D_{i,t} = D_{i,t-1}, \forall i \in \{3, 4\}$. At $\tau = 1$, the server loads the participation history $\mathcal{I}_{t-1}^1 = \{3, 4\}$, which has no intersection of the set of selected clients $\{1, 2\}$ (i.e., Condition (8) is satisfied at $\tau = 1$). Thus, the verification succeeds, and we can reuse the model $w_t^1 = w_{t-1}^1$ and the participation history $\mathcal{I}_{t-1}^1 = \mathcal{I}_t^1 = \{3, 4\}$. Next, at $\tau = 2$, the server loads the participation history $\mathcal{I}_{t-1}^2 = \{2, 3\}$, where the selected client $\{2\}$ participated in $\tau = 2$. Thus, the verification fails and the server starts retraining at $\tau = 2$ and ends at $\tau = \phi$.

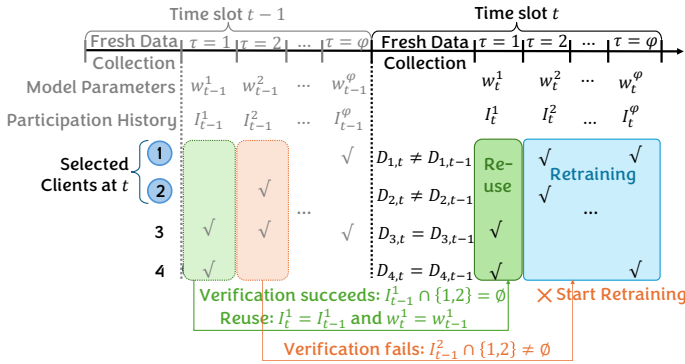


Figure 5: Illustrative Example of Algorithm 1.

APPENDIX B

CONSUMED TIME SAVINGS OF ALGORITHM 1

In this section, we characterize the *checkpoint reuse savings* of our verification-based federated unlearning algorithm in Algorithm 1 over T time slots compared to a vanilla fully-retraining approach in closed-form as follows. Let $A_{\text{Retraining}}$ be the total number of aggregations of the vanilla fully-retraining approach, and A_{Ours} be the expected total number of aggregations of our Algorithm 1 over T time slots. Since the consumed time for each aggregation is the same for vanilla fully-retraining and our Algorithm 1 if retraining is started, we define the checkpoint reuse savings as the ratio of saved aggregations (i.e., $\frac{A_{\text{Retraining}} - A_{\text{Ours}}}{A_{\text{Retraining}}}$). Let φ be the number of aggregations in each time slot, and K be the number of randomly selected clients in each aggregation as in Section IV-A.

Lemma 2. *Compared to fully-retraining, the proposed Algorithm 1 achieves a checkpoint reuse savings of:*

$$\frac{A_{\text{Retraining}} - A_{\text{Ours}}}{A_{\text{Retraining}}} = \frac{\sum_{t=2}^T R_t}{\varphi T} \geq 0, \quad (36)$$

where $R_t = \sum_{\tau=1}^{\varphi} \left(\frac{\binom{I-N_t}{K}}{\binom{I}{K}} \right)^{\tau}$ is the expected number of safely reused checkpoints (aggregation rounds) at time slot t .

Lemma 2 shows that our Algorithm 1 is strictly faster than fully-retraining. In a typical system configuration with $T = 200$, $\varphi = 5$, $I = 100$, $K = 10$, our Algorithm 1 yields a 9.78% savings in consumed time by reusing model parameters.

Proof. At any time slot $t \in [1, T]$ and aggregation round $\tau \in [1, \varphi]$, the server can safely reuse the historical checkpoint if none of the N_t currently updated clients $\mathcal{I}_t^{\text{selected}} \triangleq \{i \in \mathcal{I} | a_{i,t} = 1\}$ are present in the participation history $\mathcal{I}_{t-1}^{\tau}$ of size K . Since the K clients are sampled uniformly at random without replacement from all clients $\mathcal{I}$, the overlap size $|\mathcal{I}_t^{\text{selected}} \cap \mathcal{I}_{t-1}^{\tau}|$ follows a hypergeometric distribution [69]. Thus, the probability of successful checkpoint reuse at time slot t , aggregation τ is when the overlap size is zero:

$$P_{\text{reuse},t} = \mathbb{P}(\mathcal{I}_t^{\text{selected}} \cap \mathcal{I}_{t-1}^{\tau} = \emptyset) = \frac{\binom{I-N_t}{K}}{\binom{I}{K}}. \quad (37)$$

Due to the sequential dependence of the training process, a model is reusable only if all preceding checkpoints have also passed verification. Thus, the expected number of reusable checkpoints per time slot is:

$$R_t = \sum_{\tau=1}^{\varphi} (P_{\text{reuse},t})^{\tau} = \sum_{\tau=1}^{\varphi} \left(\frac{\binom{I-N_t}{K}}{\binom{I}{K}} \right)^{\tau}. \quad (38)$$

The total number of required aggregations for the vanilla fully-retraining approach is strictly $A_{\text{Retraining}} = \varphi T$. Conversely, our proposed mechanism requires full training only in the initial time slot ($t = 1$), and subsequently bypasses an expected R_t training rounds for all remaining slots $t \in [2, T]$. The expected total number of executed aggregations is:

$$A_{\text{Ours}} = \varphi + \sum_{t=2}^T (\varphi - R_t). \quad (39)$$

The checkpoint reuse savings ratio is derived from the fraction of bypassed aggregations:

$$\frac{A_{\text{Retraining}} - A_{\text{Ours}}}{A_{\text{Retraining}}} = \frac{\varphi T - \left(\varphi + \sum_{t=2}^T (\varphi - R_t) \right)}{\varphi T} = \frac{\sum_{t=2}^T R_t}{\varphi T}. \quad (40)$$

□

APPENDIX C

PROOF OF PROPOSITION 1

Since the unlearned model obtained by our unlearning algorithm follows the same parameter distribution as performing retraining [44], the proof follows the discussion from [47], and we extend it to an AoI-aware setting.

We first define some additional notations and derive three key lemmas in Appendix C-A, then complete the proof of Proposition 1 in Appendix C-B, and finally complete the proof of three lemmas in Appendix C-C, C-D, and C-E, respectively.

A. Additional Notations and Key Lemmas

Suppose $\nabla F_{i,t}(w_t^{\tau-1})$ is the gradient computed by all data samples in $D_{i,t}$ and $\mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{\tau-1}, x)$ is the gradient computed by a uniformly sampled subset of $D_{i,t}$.

Let

$$\bar{g}_t^\tau = \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \nabla F_{i,t}(w_t^{\tau-1}) \quad (41)$$

and

$$g_t^\tau = \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{\tau-1}, x), \quad (42)$$

we have $w_t^\tau = w_t^{\tau-1} - \eta_\tau g_t^\tau$ and $\mathbb{E} g_t^\tau = \bar{g}_t^\tau$.

Let w_t^* be the optimal model at time slot t and $w_t^{\tau-1,i}$ be the updated model computed by client i with $w_t^\tau = \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} w_t^{\tau-1,i}$.

We derive the following three key lemmas.

Lemma 3 (One Step SGD). *Assume Assumption 1 and 2, if $\eta_\tau \leq \frac{1}{4L}$, we have*

$$\begin{aligned} \mathbb{E} \|w_t^\tau - w_t^*\|^2 &\leq (1 - \eta_\tau \mu) \mathbb{E} \|w_t^{\tau-1} - w_t^*\|^2 \\ &\quad + \eta_\tau^2 \mathbb{E} \|g_t^\tau - \bar{g}_t^\tau\|^2 + 6L\eta_\tau^2 \Gamma \\ &\quad + 2\mathbb{E} \left[\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1} - w_t^{\tau-1,i}\| \right]. \end{aligned} \quad (43)$$

Proof. The proof of Lemma 3 is given in Appendix C-C. $\square$

Lemma 4 (Bounded Variance). *Assume Assumption 3, we have*

$$\mathbb{E} \|g_t^\tau - \bar{g}_t^\tau\|^2 \leq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t}^2 \sigma_i^2. \quad (44)$$

Proof. The proof of Lemma 4 is given in Appendix C-D. $\square$

Lemma 5 (Bounded Divergence). *Assume Assumption 4, η_τ is non-increasing and $\eta_\tau \leq 2\eta_{\tau+E}, \forall \tau \in [1, \varphi]$, we have*

$$\begin{aligned} &\mathbb{E} \left[\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^\tau - w_t^{\tau-1,i}\|^2 \right] \\ &\leq 4\eta_\tau^2 (E-1)^2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (G_i^2 + \Delta_{i,t} \delta_{i,t}^2) \end{aligned} \quad (45)$$

Proof. The proof of Lemma 5 is given in Appendix C-E. $\square$

B. Completing Proof of Proposition 1

Given the key lemmas, we can derive Proposition 1 as follows.

Let $\Lambda_t^\tau = \mathbb{E} \|w_t^\tau - w_t^*\|^2$. Applying Lemma 4 and Lemma 5, we can rewrite Lemma 3 as

$$\begin{aligned} \Lambda_t^\tau &\leq (1 - \eta_\tau \mu) \Lambda_t^{\tau-1} + \eta_\tau^2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t}^2 \sigma_i^2 \\ &\quad + 6L\eta_\tau^2 \Gamma + 8\eta_\tau^2 (E-1)^2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (G_i^2 + \Delta_{i,t} \delta_{i,t}^2) \\ &= (1 - \eta_\tau \mu) \Lambda_t^{\tau-1} + \eta_\tau^2 B, \end{aligned} \quad (46)$$

where B the same as $B(a_t)$ in (11).

We set up a diminishing step size as $\eta_\tau = \frac{\beta}{\tau+\gamma}$ for some $\beta > \frac{1}{\mu}$ and $\gamma > 0$ such that $\eta_1 \leq \min \left\{ \frac{1}{\mu}, \frac{1}{4L} \right\} = \frac{1}{4L}$ and

$\eta_\tau \leq \eta_{\tau+E}$. Let $v = \max \left\{ \frac{\beta^2 B}{\beta\mu-1}, (\gamma+1)\Lambda_t^1 \right\}$. We can prove $\Lambda_t^\tau \leq \frac{v}{\tau+\gamma}$ by induction as

$$\begin{aligned} \Lambda_t^\tau &\leq (1 - \eta_\tau \mu) \Lambda_t^{\tau-1} + \eta_\tau^2 B \\ &\leq \left(1 - \frac{\beta\mu}{\tau+\gamma-1}\right) \frac{v}{\tau+\gamma-1} + \frac{\beta^2 B}{(\tau+\gamma-1)^2} \\ &= \frac{\tau+\gamma-2}{(\tau+\gamma-1)^2} v + \frac{\beta^2 B}{(\tau+\gamma-1)^2} - \frac{\beta\mu-1}{(\tau+\gamma-1)^2} v \\ &\leq \frac{v}{\tau+\gamma}. \end{aligned} \quad (47)$$

By Assumption 1, we have

$$F_t(w_t) - F_t^* \leq \frac{L}{2} \Lambda_t^\varphi \leq \frac{L}{2} \frac{v}{\gamma + \varphi}. \quad (48)$$

Specifically, if we choose $\beta = \frac{2}{\mu}, \gamma = \max \left\{ \frac{8L}{\mu}, E \right\} - 1$, we have

$$\begin{aligned} v &= \max \left\{ \frac{\beta^2 B}{\beta\mu-1}, (\gamma+1)\Lambda_t^1 \right\} \\ &\leq \frac{\beta^2 B}{\beta\mu-1} + (\gamma+1)\Lambda_t^1 \\ &\leq \frac{4B}{\mu^2} + (\gamma+1)\Lambda_t^1, \end{aligned} \quad (49)$$

and therefore,

$$F_t(w_t) - F_t^* \leq \frac{L}{2} \frac{v}{\gamma + \varphi} \leq \frac{L}{2(\gamma + \varphi)} \left[\frac{4B}{\mu^2} + (\gamma+1)\Lambda_t^1 \right]. \quad (50)$$

Let $\gamma = \gamma - 1$ and apply Assumption 2, we can derive Proposition 1 as

$$F_t(w_t) - F_t^* \leq \frac{L}{2(\gamma + \varphi)} \left[\frac{4B}{\mu^2} + (\gamma+1)(F_t(w_{t-1}) - F_t^*) \right], \quad (51)$$

and we complete the proof. $\square$

C. Proof of Lemma 3

Since $w_t^\tau = w_t^{\tau-1} - \eta_\tau g_t^\tau$, we split the left-hand side of (43) into three terms as

$$\begin{aligned} &\mathbb{E} \|w_t^\tau - w_t^*\|^2 \\ &= \|w_t^{\tau-1} - \eta_\tau g_t^\tau - w_t^* - \eta_\tau \bar{g}_t^\tau + \eta_\tau \bar{g}_t^\tau\|^2 \\ &= \underbrace{\|w_t^{\tau-1} - w_t^* - \eta_\tau g_t^\tau\|^2}_{A_1} \\ &\quad + \underbrace{2\eta_\tau \langle w_t^{\tau-1} - w_t^* - \eta_\tau \bar{g}_t^\tau, \bar{g}_t^\tau - g_t^\tau \rangle}_{A_2} + \underbrace{\eta_\tau^2 \|\bar{g}_t^\tau - g_t^\tau\|^2}_{A_3}. \end{aligned} \quad (52)$$

Note that $\mathbb{E} A_2 = 0$ in (52) and A_3 is bounded by Lemma 4, we will focus on bounding A_1 for the rest of this section.

Again, we split A_1 into three terms as

$$\begin{aligned} A_1 &= \|w_t^{\tau-1} - w_t^* - \eta_\tau g_t^\tau\|^2 \\ &= \|w_t^{\tau-1} - w_t^*\|^2 - \underbrace{2\eta_\tau \langle w_t^{\tau-1} - w_t^*, \bar{g}_t^\tau \rangle}_{B_1} + \underbrace{\eta_\tau^2 \|\bar{g}_t^\tau\|^2}_{B_2}. \end{aligned} \quad (53)$$

For B_2 , we bound it by

$$B_2 = \eta_\tau^2 \|\bar{g}_t^\tau\|^2 \quad (54a)$$

$$\leq \eta_\tau^2 \left\| \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \nabla F_{i,t}(w_t^{\tau-1}) \right\|^2 \quad (54b)$$

$$\leq \eta_\tau^2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|\nabla F_{i,t}(w_t^{\tau-1})\|^2 \quad (54c)$$

$$\leq 2L\eta_\tau^2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*), \quad (54d)$$

where (54a) to (54b) is by the definition of $\bar{g}_t^\tau$ in (41), (54b) to (54c) is by the convexity of $\|\cdot\|^2$, and (54c) to (54d) is by L -smoothness of $F_{i,t}$ in Assumption 1.

For B_1 , we bound it by

$$B_1 = -2\eta_\tau \langle w_t^{\tau-1} - w_t^*, \bar{g}_t^\tau \rangle \quad (55a)$$

$$= -2\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \langle w_t^{\tau-1} - w_t^*, \nabla F_{i,t}(w_t^{\tau-1}) \rangle \quad (55b)$$

$$= -2\eta_\tau \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \langle w_t^{\tau-1} - w_t^{\tau-1,i}, \nabla F_{i,t}(w_t^{\tau-1}) \rangle}_{C_1} \\ - 2\eta_\tau \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \langle w_t^{\tau-1,i} - w_t^*, \nabla F_{i,t}(w_t^{\tau-1}) \rangle}_{C_2}, \quad (55c)$$

where (55a) to (55b) is by the definition of $\bar{g}_t^\tau$ in (41), (55b) to (55c) is by involving $w_t^{\tau-1,i}$.

By Cauchy-Schwarz inequality and AM-GM inequality, we can bound C_1 by

$$C_1 = -2\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \langle w_t^{\tau-1} - w_t^{\tau-1,i}, \nabla F_{i,t}(w_t^{\tau-1}) \rangle \\ \leq \eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \left[\frac{1}{\eta_\tau} \|w_t^{\tau-1} - w_t^{\tau-1,i}\|^2 + \eta_\tau \|\nabla F_{i,t}(w_t^{\tau-1})\|^2 \right]. \quad (56)$$

By μ -strong convexity of $F_{i,t}$, we can bound C_2 by

$$C_2 = -2\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \langle w_t^{\tau-1,i} - w_t^*, \nabla F_{i,t}(w_t^{\tau-1}) \rangle \\ \leq -2\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} [F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}(w_t^*) + \frac{\mu}{2} \|w_t^{\tau-1,i} - w_t^*\|^2]. \quad (57)$$

Then, applying the upper bound of C_1 , C_2 in B_1 , and the upper bound of B_2 back to A_1 in (53), we have

$$A_1 = \|w_t^{\tau-1} - w_t^*\|^2 - \underbrace{2\eta_\tau \langle w_t^{\tau-1} - w_t^*, \bar{g}_t^\tau \rangle}_{B_1} + \underbrace{\eta_\tau^2 \|\bar{g}_t^\tau\|^2}_{B_2} \quad (58a)$$

$$\leq \|w_t^{\tau-1} - w_t^*\|^2 + 2L\eta_\tau^2 \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*)}_{\text{Upper Bound of } B_2} \\ + \underbrace{\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \left[\frac{1}{\eta_\tau} \|w_t^{\tau-1} - w_t^{\tau-1,i}\|^2 + \eta_\tau \|\nabla F_{i,t}(w_t^{\tau-1})\|^2 \right]}_{\text{Upper Bound of } C_1} \\ - \underbrace{2\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \left[F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}(w_t^*) + \frac{\mu}{2} \|w_t^{\tau-1,i} - w_t^*\|^2 \right]}_{\text{Upper Bound of } C_2} \quad (58b)$$

$$= (1 - \mu\eta_\tau) \|w_t^{\tau-1} - w_t^*\|^2 + \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1} - w_t^{\tau-1,i}\|^2 \\ + 4L\eta_\tau^2 \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*)}_{\text{First Part of } D} \\ - 2\eta_\tau \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}(w_t^*))}_{\text{Second Part of } D}, \quad (58c)$$

where (58b) to (58c) is by reorganizing the terms, and we denote the last two terms in (58c) as D .

Next, we will bound D . Define $\gamma_\tau = 2\eta_\tau(1 - 2L\eta_\tau)$. Since $\eta_\tau \leq \frac{1}{4L}$, we have $\eta_\tau \leq \gamma_\tau \leq \eta_\tau$. Then, we can rewrite D as

$$D = -2\eta_\tau(1 - 2L\eta_\tau) \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} ((F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \\ + 2\eta_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}(w_t^*)) \\ = -\gamma_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} ((F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \\ + (2\eta_\tau - \gamma_\tau) \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}^* - F_{i,t}^*)) \\ = -\gamma_\tau \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}^*)}_E + 4L\eta_\tau^2 \Gamma. \quad (59)$$

To bound E , we have

$$E = \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}^*) \quad (60a)$$

$$= \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}(w_t^{\tau-1}) \\ + \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \quad (60b)$$

$$\geq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \langle \nabla F_{i,t}(w_t^{\tau-1}), w_t^{\tau-1,i} - w_t^{\tau-1} \rangle \\ + (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \quad (60c)$$

$$\geq -\frac{1}{2} \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} [\eta_\tau \|\nabla F_{i,t}(w_t^{\tau-1})\|^2 + \frac{1}{\eta_\tau} \|w_t^{\tau-1,i} - w_t^{\tau-1}\|^2] \\ + (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \quad (60d)$$

$$\geq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} [\eta_\tau L (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) + \frac{1}{2\eta_\tau} \|w_t^{\tau-1,i} - w_t^{\tau-1}\|^2] \\ + (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*), \quad (60e)$$

where (60a) to (60b) is by involving $F_{i,t}(w_t^{\tau-1})$, (60b) to (60c) is by the convexity of $F_{i,t}$, (60c) to (60d) is from AM-GM inequality, and (60d) to (60e) is by the L -smoothness of $F_{i,t}$.

Thus, apply the above bound of E , we can bound D by

$$D = -\gamma_\tau \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1,i}) - F_{i,t}^*)}_E + 4L\eta_\tau^2 \Gamma \\ = 4L\eta_\tau^2 \Gamma + \gamma_\tau \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} [\eta_\tau L (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \\ + \frac{1}{2\eta_\tau} \|w_t^{\tau-1,i} - w_t^{\tau-1}\|^2] - \gamma_\tau (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \\ = \gamma_\tau (\eta_\tau - 1) \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (F_{i,t}(w_t^{\tau-1}) - F_{i,t}^*) \\ + (4L\eta_\tau^2 + \gamma_\tau \eta_\tau L) \Gamma + \frac{\gamma_\tau}{2\eta_\tau} \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1,i} - w_t^{\tau-1}\|^2 \\ \leq 6L\eta_\tau^2 \Gamma + \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1,i} - w_t^{\tau-1}\|^2. \quad (61)$$

Applying the upper bound of D back to A_1 in (58c), we have

$$A_1 \leq (1 - \mu\eta_\tau) \|w_t^{\tau-1} - w_t^*\|^2 + \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1} - w_t^{\tau-1,i}\|^2 \\ + 6L\eta_\tau^2 \Gamma + \underbrace{\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1,i} - w_t^{\tau-1}\|^2}_{\text{Upper bound of } D} \\ = (1 - \mu\eta_\tau) \|w_t^{\tau-1} - w_t^*\|^2 \\ + 2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1} - w_t^{\tau-1,i}\|^2 + 6L\eta_\tau^2 \Gamma. \quad (62)$$

Finally, applying the upper bound of A_1 back into $\mathbb{E}\|w_t^\tau - w_t^*\|^2$, we have

$$\begin{aligned} \mathbb{E}\|w_t^\tau - w_t^*\|^2 &= A_1 + \underbrace{\eta_\tau^2 \|g_t^\tau - \bar{g}_t^\tau\|^2}_{A_3} \\ &\leq (1 - \mu\eta_\tau) \|w_t^{\tau-1} - w_t^*\|^2 \\ &\quad + 2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1} - w_t^{\tau-1,i}\|^2 \\ &\quad + 6L\eta_\tau^2 \Gamma + \eta_\tau^2 \|g_t^\tau - \bar{g}_t^\tau\|^2, \end{aligned} \quad (63)$$

and we complete the proof. $\square$

D. Proof of Lemma 4

By Assumption 3, we have

$$\begin{aligned} \mathbb{E}\|g_t^\tau - \bar{g}_t^\tau\|^2 &= \mathbb{E}\left\| \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (\mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{\tau-1}, x) - \nabla F_{i,t}(w_t^{\tau-1})) \right\|^2 \\ &= \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \mathbb{E} \left\| \mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{\tau-1}, x) - \nabla F_{i,t}(w_t^{\tau-1}) \right\|^2 \\ &\leq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \sigma_i^2, \end{aligned} \quad (64)$$

and we complete the proof. $\square$

E. Proof of Lemma 5

Since the local update takes E epochs, for the τ -th communication, there exists a $\tau_0 \leq \tau$ such that $\tau - \tau_0 \leq E - 1$ and $w_t^{\tau_0,i} = w_t^{\tau_0}, \forall i \in \mathcal{I}_t^{\tau_0}$. Also, we set a non-increasing learning rate η_τ with $\eta_{\tau_0} \leq 2\eta_\tau$ for all $\tau - \tau_0 \leq E - 1$.

Therefore, we have

$$\mathbb{E} \left[\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^\tau - w_t^{\tau-1,i}\|^2 \right] \quad (65a)$$

$$= \mathbb{E} \left[\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|(w_t^{\tau-1,i} - w_t^{\tau_0}) - (w_t^\tau - w_t^{\tau_0})\|^2 \right] \quad (65b)$$

$$\leq \mathbb{E} \left[\sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \|w_t^{\tau-1,i} - w_t^{\tau_0}\|^2 \right] \quad (65c)$$

$$\leq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \mathbb{E} \left[\sum_{j=t_0}^{\tau-1} (E-1)^2 \eta_j \|\mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{j-1}, x)\|^2 \right] \quad (65d)$$

$$\leq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \left[\sum_{j=t_0}^{\tau-1} (E-1)^2 \eta_j (G_i^2 + \Delta_{i,t} \delta_{i,t}^2) \right] \quad (65e)$$

$$\leq \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} \eta_{t_0}^2 (E-1)^2 (G_i^2 + \Delta_{i,t} \delta_{i,t}^2) \quad (65f)$$

$$\leq 4\eta_\tau^2 (E-1)^2 \sum_{i \in \mathcal{I}_t^\tau} h_{i,t} (G_i^2 + \Delta_{i,t} \delta_{i,t}^2) \quad (65g)$$

where (65b) to (65c) is because

$$\mathbb{E}\|X - \mathbb{E}X\|^2 \leq \mathbb{E}\|X\|^2, X = w_t^{\tau-1,i} - w_t^{\tau_0}, \quad (66)$$

(65c) to (65d) is because of Jensen inequality as

$$\begin{aligned} &\|w_t^{\tau-1,i} - w_t^{\tau_0}\|^2 \\ &= \left\| \sum_{j=t_0}^{\tau-1} \eta_j \mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{j-1}, x) \right\|^2 \\ &\leq (\tau - t_0) \sum_{j=t_0}^{\tau-1} \eta_j^2 \|\mathbb{E}_{x \in D_{i,t}} \nabla F_{i,t}(w_t^{j-1}, x)\|^2, \end{aligned} \quad (67)$$

and (65d) to (65e) is by applying Assumption 4.

We complete the proof as shown in (65g). $\square$

APPENDIX D PROOF OF LEMMA 1

Before giving the proof, we first introduce the following Chernoff-Hoeffding bound.

Lemma 6 (Chernoff-Hoeffding Bound). *Suppose that $X_1, \dots, X_n$ are n random variables with common range $[0, 1]$, satisfying $\mathbb{E}[X_t | X_1, \dots, X_{t-1}] = \mu$ for $\forall t \in [1, n]$. Let $S_n = X_1 + \dots + X_n$. Then, for all $a \geq 0$, we have*

$$\mathbb{P}[S_n \geq n\mu + a] \leq e^{-\frac{2a^2}{n}}, \text{ and } \mathbb{P}[S_n \leq n\mu - a] \leq e^{-\frac{2a^2}{n}}. \quad (68)$$

Thus, we have

$$\mathbb{P}[\hat{\delta}_{i,t} \leq \delta_i] = \mathbb{P} \left[\bar{\delta}_{i,t} + \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \leq \delta_i \right] \quad (69a)$$

$$= \mathbb{P} \left[\underbrace{n_{i,t} \bar{\delta}_{i,t}}_{S_n} + \underbrace{n_{i,t} \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}}}_a \leq \underbrace{n_{i,t} \delta_i}_{n\mu} \right] \quad (69b)$$

$$\leq e^{-2 \left[(n_{i,t} \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}})^2 / n_{i,t} \right]} \quad (69c)$$

$$= e^{-2\vartheta \ln(\sum_{j=1}^I n_{j,t})}, \quad (69d)$$

where (69a) to (69b) is by multiplying $n_{i,t}$ on both sides, (69b) to (69c) is by applying Chernoff-Hoeffding bound in Lemma 6 as labeled below the equation, and (69c) to (69d) is by organizing the terms.

On the contrary, we have

$$\mathbb{P}[0 < \hat{\delta}_{i,t} - \delta_i] \geq 1 - e^{-2\vartheta \ln(\sum_{j=1}^I n_{j,t})}. \quad (70)$$

Similar to (69), we have

$$\mathbb{P} \left[\hat{\delta}_{i,t} - 2\sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \geq \delta_i \right] \quad (71a)$$

$$= \mathbb{P} \left[\bar{\delta}_{i,t} + \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} - 2\sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \geq \delta_i \right] \quad (71b)$$

$$= \mathbb{P} \left[n_{i,t} \bar{\delta}_{i,t} - n_{i,t} \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \geq n_{i,t} \delta_i \right] \quad (71c)$$

$$= \mathbb{P} \left[\underbrace{n_{i,t} \bar{\delta}_{i,t}}_{S_n} \geq \underbrace{n_{i,t} \delta_i}_{n\mu} + \underbrace{n_{i,t} \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}}}_a \right] \quad (71d)$$

$$\leq e^{-2 \left[(n_{i,t} \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}})^2 / n_{i,t} \right]} \quad (71e)$$

$$= e^{-2\vartheta \ln(\sum_{j=1}^I n_{j,t})}, \quad (71f)$$

On the contrary, we have □

$$\mathbb{P} \left[\hat{\delta}_{i,t} - \delta_i < 2\sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \right] \geq 1 - e^{-2\vartheta \ln(\sum_{j=1}^I n_{j,t})}. \quad (72)$$

Combining (70) and (72), we have

$$\mathbb{P} \left[0 < \hat{\delta}_{i,t} - \delta_i < 2\sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \right] \geq \left(1 - e^{-2\vartheta \ln(\sum_{j=1}^I n_{j,t})} \right)^2 \quad (73)$$

Then, when $t \rightarrow \infty$, we have

$$\begin{aligned} \lim_{t \rightarrow \infty} \mathbb{P} \left[0 < \hat{\delta}_{i,t} - \delta_i < 2\sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \right] \\ \geq \lim_{t \rightarrow \infty} \left(1 - e^{-2\vartheta \ln(\sum_{j=1}^I n_{j,t})} \right)^2 = 1, \end{aligned} \quad (74)$$

and we complete the proof. □

APPENDIX E PROOF OF PROPOSITION 2

A. Proof of Truthfulness

Let the selection of client i by reporting true data collection cost $c_{i,t}$ and non-truthful cost $b_{i,t}$ be $a_{i,t}$ and $a'_{i,t}$, respectively. There are four possible combinations $(a_{i,t}, a'_{i,t}) \in \{(1, 1), (1, 0), (0, 1), (0, 0)\}$. Next, we discuss two cases separately.

Case 1 ($b_{i,t} > c_{i,t}$). In Case 1, we prove that reporting a higher data collection cost cannot increase the utility by analyzing all four combinations of $(a_{i,t}, a'_{i,t})$. Firstly, if $(a_{i,t}, a'_{i,t}) = (1, 1)$, $p_{i,t}$ will not change according to our payment rule in (30). Secondly, $(a_{i,t}, a'_{i,t}) = (1, 0)$, it means client i is selected if honestly reporting. That is, $\frac{\hat{\theta}_{i,t} - c_{i,t}}{d_i} \geq \hat{g}_t(c_{i,t}, \mathbf{b}_{-i,t})[N_t + 1]$ is satisfied, which is equivalent to $\hat{\theta}_{i,t} - d_i \hat{g}_t(c_{i,t}, \mathbf{b}_{-i,t})[N_t + 1] - c_{i,t} = p_{i,t} - c_{i,t} \geq 0$ according to our payment rule in (30). It shows that client i can obtain a non-negative utility when honestly reporting. However, it can only obtain zero utility by non-truthfully reporting because it is not selected (i.e., $a'_{i,t} = 0$). Thus, in this case, non-truthfulness cannot bring higher utility. Thirdly, $(a_{i,t}, a'_{i,t}) = (0, 1)$ is impossible because reporting a higher cost only decreases $\hat{g}_{i,t}$ and thus decreases the chance for being selected. Lastly, the utility in $(a_{i,t}, a'_{i,t}) = (0, 0)$ is the same (i.e., $u_{i,t} = 0$) by misreporting.

Case 2 ($b_{i,t} < c_{i,t}$). In Case 2, we prove that reporting a lower data collection cost cannot increase the utility by analyzing all four combinations of $(a_{i,t}, a'_{i,t})$. Firstly, if $(a_{i,t}, a'_{i,t}) = (1, 1)$, $p_{i,t}$ will not change according to our payment rule in (30). Secondly, $(a_{i,t}, a'_{i,t}) = (1, 0)$ is impossible because reporting a lower cost only increases $\hat{g}_{i,t}$ and thus increases the chance for being selected. Thirdly, $(a_{i,t}, a'_{i,t}) = (0, 1)$ means client i cannot be selected if honestly reports. That is, $\frac{\hat{\theta}_{i,t} - c_{i,t}}{d_i} \leq \hat{g}_t(c_{i,t}, \mathbf{b}_{-i,t})[N_t + 1]$. Applying this condition, we can compute its utility after reporting a lower cost $b_{i,t}$ by $u_{i,t} = p_{i,t} - c_{i,t} \leq 0$. That means reporting a lower data collection cost results in a negative utility. Lastly, the utility in $(a_{i,t}, a'_{i,t}) = (0, 0)$ is the same (i.e., $u_{i,t} = 0$) by misreporting.

B. Proof of Individual Rationality

If client i is selected, that means $\frac{\hat{\theta}_{i,t} - c_{i,t}}{d_i} \geq \hat{g}_t(\mathbf{b}_t)[N_t + 1]$ is satisfied, which is equivalent to $\hat{\theta}_{i,t} - d_i \hat{g}_t(\mathbf{b}_t)[N_t + 1] - c_{i,t} = p_{i,t}(\mathbf{b}_t) - c_{i,t} \geq 0$. It shows that client i can obtain a non-negative utility if selected. Otherwise, when client i is not selected, its utility is zero. □

C. Proof of Computational Efficiency

For client selection rule in Algorithm 2, the computational time includes:

- (line 2) the time for computing $\hat{\theta}_{i,t}$ and $\hat{g}_{i,t}$ in $\mathcal{O}(I)$
- (line 3) the time for sorting client's $\hat{g}_{i,t}$ in $\mathcal{O}(I \log I)$ by Quick Sort [70]
- (line 4-8) the time for determining client selection in $\mathcal{O}(I)$

For the payment rule in (30), it takes $\mathcal{O}(I)$ to determine the payments.

Therefore, the server can compute $a_{i,t}(\mathbf{b}_t)$ and $p_{i,t}(\mathbf{b}_t)$ in $\mathcal{O}(I)$. □

APPENDIX F PROOF OF PROPOSITION 3

For clarity, we denote our client selection policy in Algorithm 3 as $\{\mathbf{a}_t^*\}_{\forall t \in \mathcal{T}}$ in this appendix. The corresponding server's long-term average cost achieved by our policy is

$$\phi = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right]. \quad (75)$$

Let the offline optimal client selection policy be $\{\mathbf{a}_t^o\}_{\forall t \in \mathcal{T}}$ and the corresponding optimal server's long-term average cost be

$$\phi_{\min} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_t^o)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^o p_{i,t} \right]. \quad (76)$$

Next, let the upper bound of the drift-plus-penalty term obtained by $\mathbf{a}_t^*$ and $\mathbf{a}_t^o$ at time slot t be

$$\begin{aligned} \phi_t^* = & C + \sum_{i=1}^I Q_i(t) \left(a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \right) \\ & + V \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right], \end{aligned} \quad (77)$$

and

$$\begin{aligned} \phi_t^o = & C + \sum_{i=1}^I Q_i(t) \left(a_{i,t}^o - \frac{M}{\sum_{i=1}^I d_i} \right) \\ & + V \left[\frac{2LB(\mathbf{a}_t^o)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^o p_{i,t} \right], \end{aligned} \quad (78)$$

respectively.

Further, suppose there is a special policy $\{\xi_t\}_{\forall t \in \mathcal{T}}$ that satisfies its obtained server's cost at each time slot equals the minimum long-term average server's cost $\phi_{\min}$, as

$$\frac{2LB(\xi_t)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I \xi_{i,t} p_{i,t} = \phi_{\min}, \forall t \in \mathcal{T}. \quad (79)$$

Therefore, we have

$$\phi_t^o \leq C + \sum_{i=1}^I Q_i(t) \left(\xi_{i,t} - \frac{M}{\sum_{i=1}^I d_i} \right) + V \left[\frac{2LB(\xi_t)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I \xi_{i,t} p_{i,t} \right] \quad (80a)$$

$$\leq C + V\phi_{\min} + \sum_{i=1}^I Q_i(t) \left(\xi_{i,t} - \frac{M}{\sum_{i=1}^I d_i} \right), \quad (80b)$$

where (80a) is because ϕ_t^o , the drift-plus-penalty term obtained by the offline optimal policy $\mathbf{a}_t^o$, should not be larger than the drift-plus-penalty term obtained by any other policy (e.g., ξ_t), and (80a) to (80b) is because of (79).

Then, we can find that the drift-plus-penalty term by our policy $\mathbf{a}_t^*$ satisfies

$$\Delta(\Theta(t)) + V \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \quad (81a)$$

$$\leq C + \sum_{i=1}^I Q_i(t) \left(a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \right) + V \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \quad (81b)$$

$$= \phi_t^* \quad (81c)$$

$$\leq \phi_t^g + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \mu) \sum_{i=1}^I d_i} d^{\max}(\Delta_{t-1}^{\max} + k^c) N_t \epsilon_t \quad (81d)$$

$$\leq 2\phi_t^o + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \mu) \sum_{i=1}^I d_i} d^{\max}(\Delta_{t-1}^{\max} + k^c) N_t \epsilon_t \quad (81e)$$

$$\leq 2 \left(C + V\phi_{\min} + \sum_{i=1}^I Q_i(t) \left(\xi_{i,t} - \frac{M}{\sum_{i=1}^I d_i} \right) \right) + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \mu) \sum_{i=1}^I d_i} d^{\max}(\Delta_{t-1}^{\max} + k^c) N_t \epsilon_t, \quad (81f)$$

where (81a) to (81b) is because of the upper bound of Lyapunov drift in (17), (81b) to (81c) is because of the definition in (77), (81c) to (81d) is because of Lemma 7, (81d) to (81e) is because the greedy selection when server knows $\delta_{i,t}$ and $c_{i,t}$ achieves $\frac{1}{2}$ -approximation [71] of the offline optimal solution ϕ_t^o , and (81e) to (81f) is because of (80).

Lemma 7. Suppose there is a server that has complete knowledge of clients' private data criticality $\theta_{i,t}$ and data collection cost $c_{i,t}$. The server adopts a policy $\{\mathbf{a}_t^g\}$ that greedily selects clients with the largest $g_{i,t} = \frac{\theta_{i,t} - c_{i,t}}{d_i}$. Let the drift-plus-penalty term obtained by the server with complete knowledge at time slot t be ϕ_t^g . The gap between ϕ_t^g and ϕ_t^* is bounded by

$$\phi_t^* \leq \phi_t^g + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \phi) \sum_{i=1}^I d_i} d^{\max} \Delta_{t-1}^{\max} N_t \epsilon_t. \quad (82)$$

Proof. Let $\mathcal{I}_t^S$ and $\mathcal{I}_t^A$ be the set of selected clients by our policy $\{\mathbf{a}_t^*\}$ and $\{\mathbf{a}_t^g\}$. Let $\zeta_{i,t} = d_i(\Delta_{i,t-1} + k^c)$ and $\zeta_t^{\max} =$

$\max\{\zeta_{1,t}, \dots, \zeta_{I,t}\}$. Let $\epsilon_t = \max\{\hat{\delta}_{i,t}^2 - \delta_{i,t}^2\}_{\forall i \in \mathcal{I}}$ be the UCB estimation error. Thus, we have

$$\sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} \hat{\delta}_{i,t}^2 - \sum_{i \in \mathcal{I}_t^A} \zeta_{i,t} \delta_{i,t}^2 \quad (83a)$$

$$\leq \sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} (\hat{\delta}_{i,t}^2 - \delta_{i,t}^2) + \sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} \delta_{i,t}^2 - \sum_{i \in \mathcal{I}_t^A} \zeta_{i,t} \delta_{i,t}^2 \quad (83b)$$

$$\leq \zeta_t^{\max} N_t \epsilon_t, \quad (83c)$$

where (83a) to (83b) involves a term $\sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} \delta_{i,t}^2$, (83b) to (83c) is because policy $\{\mathbf{a}_t^g\}$ can maximize the sum of selected clients' $\zeta_{i,t} \delta_{i,t}^2$ and $\sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} \delta_{i,t}^2 - \sum_{i \in \mathcal{I}_t^A} \zeta_{i,t} \delta_{i,t}^2 \leq 0$.

By multiplying term $\Psi = \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \phi) \sum_{i=1}^I d_i}$ on both sides, we have

$$\Psi \left(\sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} \hat{\delta}_{i,t}^2 - \sum_{i \in \mathcal{I}_t^A} \zeta_{i,t} \delta_{i,t}^2 \right) \leq \Psi \zeta_t^{\max} N_t \epsilon_t. \quad (84)$$

Thus, we have

$$\begin{aligned} & \Psi \sum_{i \in \mathcal{I}_t^S} \zeta_{i,t} \hat{\delta}_{i,t}^2 - \sum_{i \in \mathcal{I}_t^S} \frac{Q_i(t)}{V\alpha} - \sum_{i \in \mathcal{I}_t^S} c_{i,t} \\ & \leq \Psi \zeta_t^{\max} N_t \epsilon_t + \Psi \sum_{i \in \mathcal{I}_t^A} \zeta_{i,t} \delta_{i,t}^2 \\ & \quad - \sum_{i \in \mathcal{I}_t^A} \frac{Q_i(t)}{V\alpha} - \sum_{i \in \mathcal{I}_t^A} c_{i,t}, \end{aligned} \quad (85)$$

which is equivalent to

$$\sum_{i \in \mathcal{I}_t^S} (\hat{\theta}_{i,t} - c_{i,t}) \leq \sum_{i \in \mathcal{I}_t^A} (\theta_{i,t} - c_{i,t}) + \Psi \zeta_t^{\max} N_t \epsilon_t. \quad (86)$$

Given Proposition 2 that ensures $b_{i,t} = c_{i,t}$, we have

$$\phi_t^* \leq \phi_t^g + \Psi \zeta_t^{\max} N_t \epsilon_t, \quad (87)$$

which can rewrite as

$$\phi_t^* \leq \phi_t^g + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \phi) \sum_{i=1}^I d_i} d^{\max} \Delta_{t-1}^{\max} N_t \epsilon_t, \quad (88)$$

and we complete the proof. $\square$

By summing up T time slots of (81a) and (81f), we have

$$\begin{aligned} & \sum_{t=1}^T \Delta(\Theta(t)) + V \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \\ & \leq 2T(C + V\phi_{\min}) + 2 \sum_{t=1}^T \sum_{i=1}^I Q_i(t) \left(\xi_{i,t} - \frac{M}{\sum_{i=1}^I d_i} \right) \\ & \quad + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma + \mu) \sum_{i=1}^I d_i} \sum_{t=1}^T d^{\max}(\Delta_{t-1}^{\max} + k^c) N_t \epsilon_t. \end{aligned}$$

On the other side, we have

$$\sum_{t=1}^T \Delta(\Theta(t)) + V \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \quad (89a)$$

$$\begin{aligned} & = \sum_{t=1}^T [L(\Theta(t+1)) - L(\Theta(t))] \\ & \quad + V \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \end{aligned} \quad (89b)$$

$$\geq L(\Theta(T)) - L(\Theta(0)) + V \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \quad (89c)$$

$$= \frac{1}{2} \sum_{i=1}^I Q_i(T)^2 + V \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right], \quad (89d)$$

where (89a) to (89b) is because of the definition in (16), and (89c) to (89d) is because of the definition in (15).

Combining (89) and (89d), we have

$$\begin{aligned} & \frac{1}{2} \sum_{t=1}^T Q_i(T)^2 + V \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_i^*)}{\mu^2(\gamma+\varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \\ & \leq 2T(C + V\phi_{\min}) + 2 \sum_{t=1}^T \sum_{i=1}^I Q_i(t) \left(\xi_{i,t} - \frac{M}{\sum_{i=1}^I d_i} \right) \\ & \quad + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma+\mu) \sum_{i=1}^I d_i} \sum_{t=1}^T d^{\max}(\Delta_{t-1}^{\max} + k^c) N_t \epsilon_t. \end{aligned} \quad (90)$$

Next, we prove $\lim_{T \rightarrow \infty} \sum_{i=1}^I Q_i(T) = 0$. In each time slot, we have

$$\begin{aligned} Q_i(t) & \leq Q_i(t-1) + a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \\ & \leq Q_i(t-2) + a_{i,t-1}^* + a_{i,t}^* - \frac{2M}{\sum_{i=1}^I d_i} \\ & \leq \dots \\ & \leq Q_i(0) + \sum_{j=1}^t a_{i,j}^* - \frac{tM}{\sum_{i=1}^I d_i} \\ & = \sum_{j=1}^t a_{i,j}^* - \frac{tM}{\sum_{i=1}^I d_i}. \end{aligned} \quad (91)$$

Summing up for each client, we have

$$\sum_{i=1}^I Q_i(t) \leq \sum_{i=1}^I \sum_{j=1}^t a_{i,j}^* - \frac{ItM}{\sum_{i=1}^I d_i}. \quad (92)$$

When $t \rightarrow \infty$, we have $\frac{1}{t} \sum_{j=1}^t a_{i,j}^* \leq \frac{M}{\sum_{i=1}^I d_i}$ because of the fairness constraint (5) and we prove the fairness constraint is satisfied in Appendix G. Therefore, we have $\lim_{t \rightarrow \infty} \sum_{j=1}^t a_{i,j}^* \leq \frac{tM}{\sum_{i=1}^I d_i}$.

Thus, we can prove that, when $t \rightarrow \infty$,

$$\begin{aligned} \sum_{i=1}^I Q_i(t) & \leq \sum_{i=1}^I \frac{tM}{\sum_{i=1}^I d_i} - \frac{ItM}{\sum_{i=1}^I d_i} \\ & = \frac{ItM}{\sum_{i=1}^I d_i} - \frac{ItM}{\sum_{i=1}^I d_i} = 0. \end{aligned} \quad (93)$$

Applying (93) to (90), when $T \rightarrow \infty$, we have

$$\begin{aligned} & V \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_i^*)}{\mu^2(\gamma+\varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \\ & \leq 2T(C + V\phi_{\min}) \\ & \quad + \frac{16L(E-1)^2}{\alpha\mu^2(\gamma+\mu) \sum_{i=1}^I d_i} \sum_{t=1}^T d^{\max}(\Delta_{t-1}^{\max} + k^c) N_t \epsilon_t, \end{aligned} \quad (94)$$

which is equivalent to

$$\begin{aligned} \phi & = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_i^*)}{\mu^2(\gamma+\varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \\ & \leq \frac{2(C+V\phi_{\min})}{V} + \frac{16L(E-1)^2 I d^{\max}}{V\alpha\mu^2(\gamma+\mu) \sum_{i=1}^I d_i} \frac{1}{T} \sum_{t=1}^T (\Delta_{t-1}^{\max} + k^c) \epsilon_t. \end{aligned} \quad (95)$$

Finally, we prove the value of $\frac{1}{T} \sum_{t=1}^T (\Delta_{t-1}^{\max} + k^c) \epsilon_t$ is sub-linear to T . Because of fairness in Proposition 4 and the definition of $\Delta_{i,t-1}$ in (2), we have $\Delta_{t-1}^{\max} \leq \sum_{\tau=1}^t k_{\tau}^f + k^c \frac{(1 - \frac{M}{\sum_{i=1}^I d_i}) - (1 - \frac{M}{\sum_{i=1}^I d_i})^t}{\frac{M}{\sum_{i=1}^I d_i}}$.

For ϵ_t , we have

$$\epsilon_t = \underbrace{\bar{\delta}_{i,t} - \delta_i}_{\text{Sample Mean Error}} + \underbrace{\delta_i - \delta_{i,t}}_{\text{One Time Error}} + \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}}. \quad (96)$$

We bound the sample mean error by the Chernoff-Hoeffding bound in Lemma 6 as

$$\mathbb{P} \left[\bar{\delta}_{i,t} - \delta_i \geq \sqrt{\frac{u}{2n_{i,t}}} \right] \leq e^{-u}, \forall u \geq 0. \quad (97)$$

Since $|\delta_{i,t} - \delta_i|$ follows ω -sub-gaussian distribution, let $\psi = \omega\sqrt{2u}$ with any $u \geq 0$ and apply the definition in (31), we have

$$\mathbb{P}[|\delta_{i,t} - \delta_i| \geq \omega\sqrt{2u}] \leq 2e^{-u}, \forall u \geq 0. \quad (98)$$

By the symmetry of sub-gaussian distribution, we have

$$\mathbb{P}[\delta_{i,t} - \delta_i \geq \omega\sqrt{2u}] \leq e^{-u}, \forall u \geq 0. \quad (99)$$

Let $u = \ln \sum_{j=1}^I n_{j,t}$, there is a probability greater than $1 - \frac{2}{\sum_{j=1}^I n_{j,t}}$ that

$$\begin{aligned} \epsilon_t & \leq \omega \sqrt{2 \ln \sum_{j=1}^I n_{j,t}} + \sqrt{\frac{\ln \sum_{j=1}^I n_{j,t}}{2n_{i,t}}} + \sqrt{\frac{\vartheta \ln(\sum_{j=1}^I n_{j,t})}{n_{i,t}}} \\ & \leq \sqrt{2 \ln \sum_{j=1}^I n_{j,t} \left(\frac{2\vartheta+1}{n_{i,t}} + 2\omega^2 \right)}. \end{aligned} \quad (100)$$

When $T \rightarrow \infty$, we have

$$\begin{aligned} \mathbb{P} \left[\epsilon_t \leq \sqrt{2 \ln \frac{MI}{\sum_{i=1}^I d_i} \left(\frac{(2\vartheta+1) \sum_{i=1}^I d_i}{M} + 2\omega^2 \right)} \right] \\ \geq 1 - \frac{2}{\sum_{j=1}^I n_{j,t}} \rightarrow 1. \end{aligned} \quad (101)$$

Therefore, we have

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T (\Delta_{t-1}^{\max} + k^c) \epsilon_t \\ & \leq \frac{1}{T} \sum_{t=1}^T \left[\sum_{\tau=1}^t k_{\tau}^f + k^c \frac{(1 - \frac{M}{\sum_{i=1}^I d_i}) - (1 - \frac{M}{\sum_{i=1}^I d_i})^t}{\frac{M}{\sum_{i=1}^I d_i}} \right] \\ & \quad \left(\sqrt{2(2\vartheta+1) \ln \frac{IM}{\sum_{i=1}^I d_i} \left(\frac{\sum_{i=1}^I d_i}{M} \right)} + \sqrt{2 \ln \frac{IM}{\sum_{i=1}^I d_i} 2\omega^2} \right) \\ & \leq \left(\frac{1}{T} \sum_{t=1}^T \sum_{\tau=1}^t k_{\tau}^f + k^c \right) \\ & \quad \sqrt{16(2\vartheta+1) \frac{\sum_{i=1}^I d_i}{M} \omega^2 \ln \frac{IM}{\sum_{i=1}^I d_i} \int_1^T \sqrt{t} dt} \\ & \leq \frac{2}{3} \left(\frac{1}{T} \sum_{t=1}^T \sum_{\tau=1}^t k_{\tau}^f + k^c \right) \\ & \quad \sqrt{16(2\vartheta+1) \frac{\sum_{i=1}^I d_i}{M} \omega^2 \ln \frac{IM}{\sum_{i=1}^I d_i} \sqrt{T}} \\ & = \mathcal{O}(\omega\sqrt{T}), \end{aligned} \quad (102)$$

with a probability greater than $1 - \frac{2}{IT}$. Let $\Theta = \frac{16L(E-1)^2 I d^{\max}}{V\alpha\mu^2(\gamma+\mu) \sum_{i=1}^I d_i} \frac{1}{T} \sum_{t=1}^T (\Delta_{t-1}^{\max} + k^c) \epsilon_t = \mathcal{O}(\omega\sqrt{T})$ and $\Omega = 1 - \frac{2}{IT}$. Applying (95), we have

$$\begin{aligned} \phi & = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left[\frac{2LB(\mathbf{a}_i^*)}{\mu^2(\gamma+\varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \\ & \leq \frac{2(C+V\phi_{\min})}{V} + \Theta, \end{aligned} \quad (103)$$

with a probability greater than Ω and we complete the proof. $\square$

APPENDIX G PROOF OF PROPOSITION 4

For clarity, we denote our client selection policy in Algorithm 3 as $\{\mathbf{a}_t^*\}_{t \in \mathcal{T}}$ in this appendix. The drift-plus-penalty term obtained by our policy in Algorithm 3 satisfies

$$\Delta(\Theta(t)) + V\phi_{\min} \quad (104a)$$

$$\leq \Delta(\Theta(t)) + V \left[\frac{2LB(\mathbf{a}_t^*)}{\mu^2(\gamma + \varphi)} + \alpha \sum_{i=1}^I a_{i,t}^* p_{i,t} \right] \quad (104b)$$

$$\leq C + V\phi_{\max} + \sum_{i=1}^I Q_i(t) \left(a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \right) \quad (104c)$$

where (104a) to (104b) is because the offline optimal policy always achieve the lowest server's cost than any other policy (i.e., $\mathbf{a}_t^*$ here), and (104b) to (104c) is because $\phi_{\max}$ is the largest server cost and the upper bound of $\Delta(\Theta(t))$ in (17).

Therefore, we have

$$\Delta(\Theta(t)) = L(\Theta(t+1)) - L(\Theta(t)) \quad (105a)$$

$$\leq C + V(\phi_{\max} - \phi_{\min}) + \sum_{i=1}^I Q_i(t) \left(a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \right) \quad (105b)$$

where (105a) to (105b) is by moving terms in (104a) and (104c).

By summing up T time slots in (105), when $T \rightarrow \infty$, we have

$$\sum_{t=1}^T \Delta(\Theta(t)) = L(\Theta(T)) - L(\Theta(0)) = \frac{1}{2} \sum_{i=1}^I Q_i(T)^2 \quad (106a)$$

$$\leq T[C + V(\phi_{\max} - \phi_{\min})] + \sum_{t=1}^T \sum_{i=1}^I Q_i(t) \left(a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \right) \quad (106b)$$

$$\leq T[C + V(\phi_{\max} - \phi_{\min})], \quad (106c)$$

where (106a) is because of the definitions in (16) and (15), and (106b) to (106c) is because of (93).

By Cauchy-Schwarz inequality, we have

$$\left[\sum_{i=1}^I Q_i(T) \right]^2 \leq I \sum_{i=1}^I Q_i(T)^2 \leq 2IT[C + V(\phi_{\max} - \phi_{\min})], \quad (107)$$

and therefore,

$$\sum_{i=1}^I Q_i(T) \leq \sqrt{2IT[C + V(\phi_{\max} - \phi_{\min})]}. \quad (108)$$

Dividing (108) by T , when $T \rightarrow \infty$, we have

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{i=1}^I Q_i(T) \leq \sqrt{\frac{2I[C + V(\phi_{\max} - \phi_{\min})]}{T}} = 0. \quad (109)$$

Because $Q_i(T) \leq \sum_{i=1}^I Q_i(T)$, we have

$$\lim_{T \rightarrow \infty} \frac{Q_i(T)}{T} \leq \sqrt{\frac{2I[C + V(\phi_{\max} - \phi_{\min})]}{T}} = 0, \forall i \in \mathcal{I}. \quad (110)$$

Given the definition in (13), we have

$$Q_i(t+1) \geq Q_i(t) + a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i}, \quad (111)$$

which is equivalent to

$$a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} \leq Q_i(t+1) - Q_i(t). \quad (112)$$

By summing up (112) in T time slots, we have

$$\begin{aligned} \sum_{t=1}^T a_{i,t}^* - \frac{TM}{\sum_{i=1}^I d_i} &\leq \sum_{t=1}^T [Q_i(t+1) - Q_i(t)] \\ &= Q_i(T) - Q_i(0) = Q_i(T). \end{aligned} \quad (113)$$

Dividing (113) by T on both sides and applying (110), when $T \rightarrow \infty$, we can get

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T a_{i,t}^* - \frac{M}{\sum_{i=1}^I d_i} &\leq \lim_{T \rightarrow \infty} \frac{1}{T} Q_i(T) \\ &\leq \sqrt{\frac{2I[C + V(\phi_{\max} - \phi_{\min})]}{T}} = 0, \forall i \in \mathcal{I}, \end{aligned} \quad (114)$$

and we complete the proof. $\square$

APPENDIX H EXPERIMENTS RESULTS ON BASELINES COMPARISON

In this section, we demonstrate the performance of MNIST [24] dataset under Noise in Figure 6, Cifar-100 [25] dataset under Mislabeling in Figure 7, Cifar-100 [25] dataset under Noise in Figure 8, and CICIOMT [62] dataset in Figure 9. In each figure, we have three sub-figures that show the server's cost, validation accuracy, and total payment, respectively. Note that we show the Top-5 validation accuracy in Cifar-100 [25] dataset because it contains a large number of classes. For CICIOMT [62] dataset, we set the scaling parameters as $V = 1$ and $\alpha = 0.5$. We set $\varphi = 2$ and the data collection time as $k^c = 1$. The other parameter settings are the same as Section VII-A. From the results, we can find that our Observation 1 is still valid in different datasets and data modifications.

APPENDIX I IMPACT OF SCALING PARAMETER α

In this section, to investigate the impact of the scaling parameter between model's loss and total payment in server's cost in (3), we set three representative values as $\alpha = 1, 10, 100$. The results on MNIST [24] dataset under Mislabeling are shown in Figure 10.

In Figure 10a, we compare the model's validation accuracy under various α , as a more intuitive metric of model performance rather than model's loss optimality gap. We find that a larger α results in a lower accuracy. Since the accuracy is not stable, we compare the average accuracy in the last 20 time slots instead. We find that the average accuracy of $\alpha = 1$ is 8.49% and 19.31% higher than $\alpha = 10$ and $\alpha = 100$ in the last 20 time slots. The reason is that a larger α in (3) emphasizes minimizing the payment rather than the model's loss. The server selects some low-cost clients to save the payment but sacrifices some optimality in model's loss.

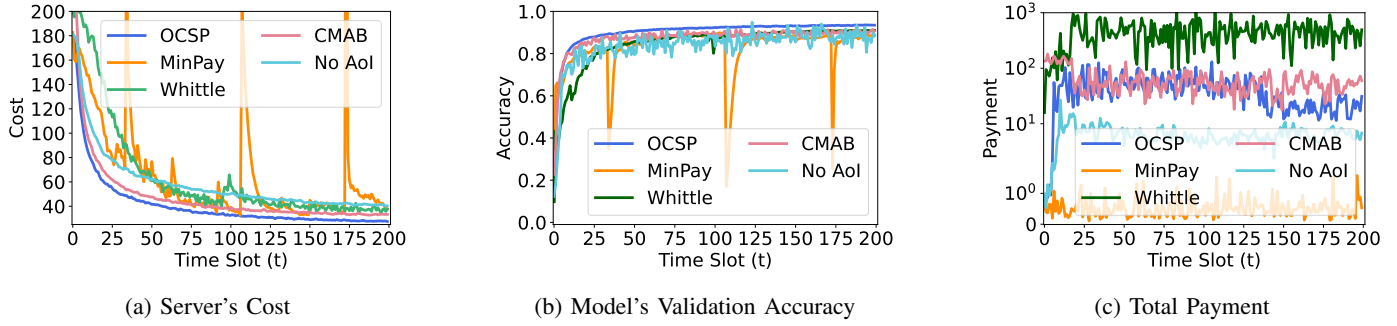


Figure 6: Performance of MNIST dataset Under Noise

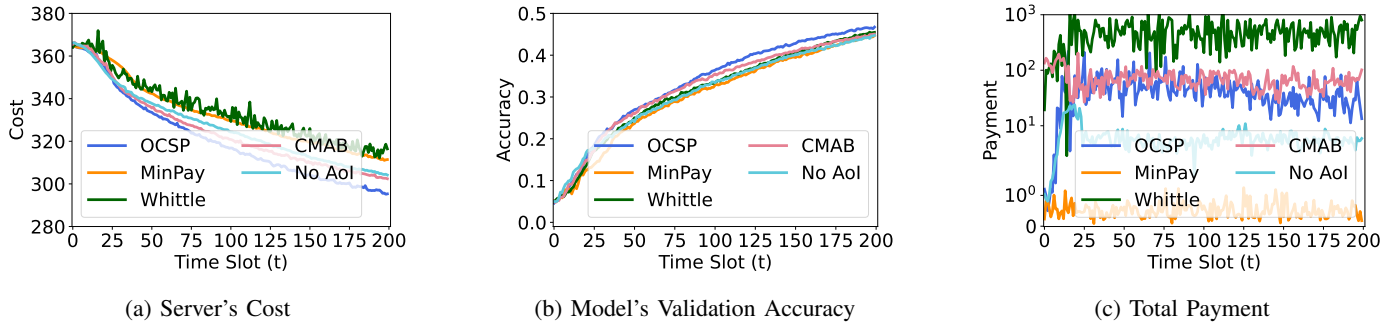


Figure 7: Performance of Cifar-100 dataset Under Mislabeling

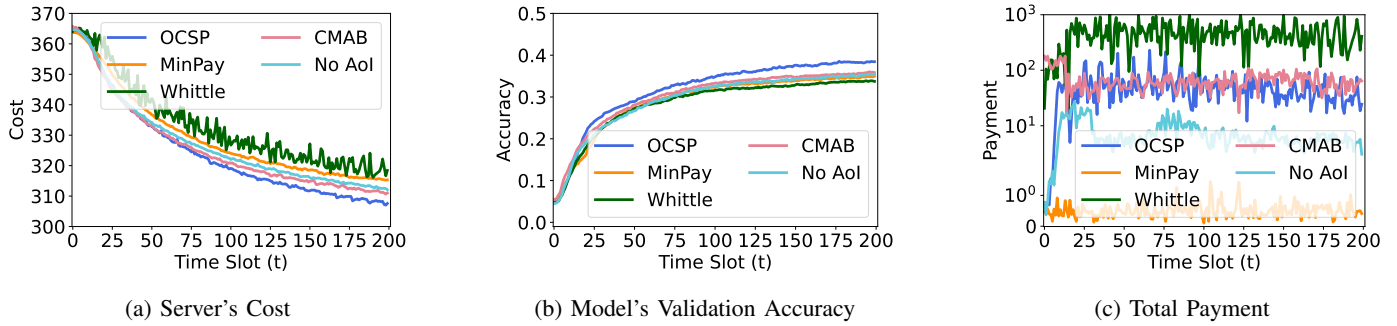


Figure 8: Performance of Cifar-100 dataset Under Noise

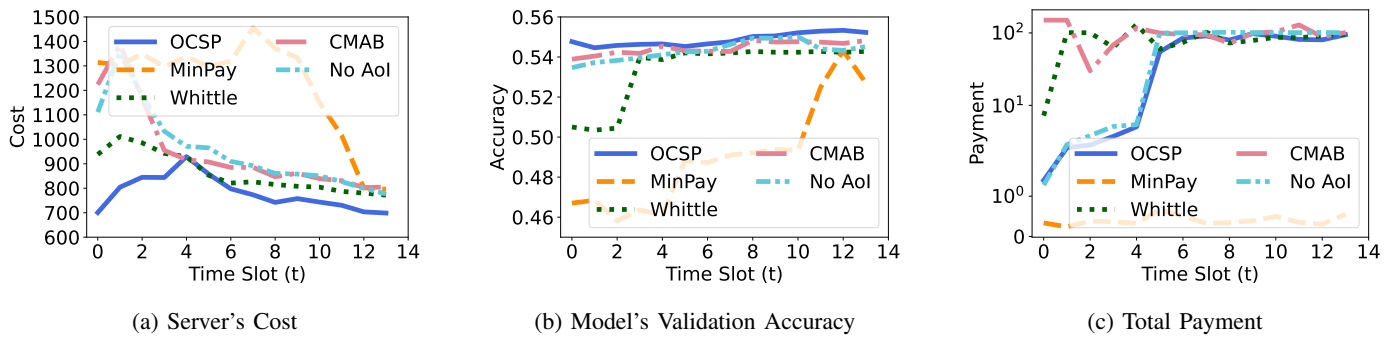


Figure 9: Performance of CICIoMT dataset

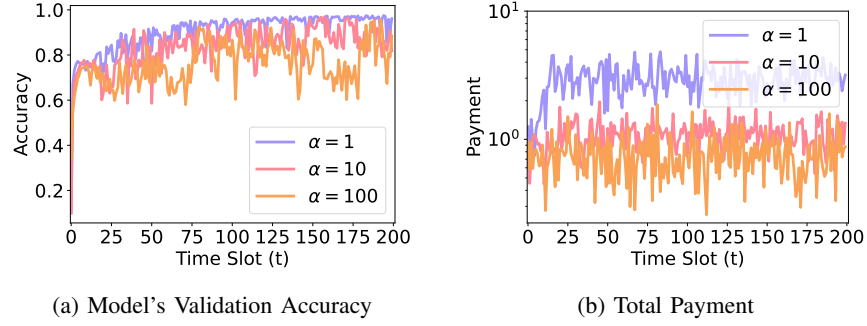


Figure 10: Performance of Various Scaling Parameter α under OCSF mechanism

In Figure 10b, we show the total payment for different values of α . We observe that the payment decreases with α . Similarly, we compare the average payment in the last 20 time slots and find that the payment of $\alpha = 10$ and $\alpha = 100$ is 64.33% and 76.43% smaller than $\alpha = 1$. The reason is the same as Figure 10a that a larger α minimizes the payment over model's loss.

We summarize our key observation as follows.

Observation 4. A larger α reduces the payment over the model's loss optimality.

APPENDIX J IMPACT OF NON-IID LEVEL

We use the sharding approach [46] to distribute the dataset. It sorts the data by label, chops it into distinct shards, and distributes a random number of shards to each client in order. This forces clients to have only 1 or 2 specific labels. It cannot be directly extended to different non-IID levels. Thus, we adopt the commonly used Dirichlet distribution [72] (i.e., $Dir(\chi)$) to simulate different levels of non-IID in FL. We set up three different non-IID levels with $\chi = 0.2, 0.5, 1$. A larger χ results in more similar label distributions across clients, whereas $\chi \rightarrow 0$ means each client holds data from only a single class.

We use MNIST [24] dataset under Mislabeling as an example to demonstrate the impact of χ . We keep all parameter settings the same. The results are shown in Figure 11.

In Figure 11a, we compare the server's cost under different χ . We find that the server's cost decreases with χ . Specifically, $\chi = 1$ improves the server's cost by 4.84% and 10.07% compared with $\chi = 0.5$ and $\chi = 0.2$ in the last 20 slots (after the costs have converged) on average. It is because a larger χ improves the unlearned model's accuracy while not increasing the payment, which we demonstrate in Figure 11b and Figure 11c.

In Figure 11b, we compare the accuracy on the validation set. We find that $\chi = 1$ achieves the highest model accuracy. It is because a larger Dirichlet parameter χ leads to a more balanced data distribution, reduces local data heterogeneity, and minimizes model aggregation divergence. It improves overall accuracy and also makes the global model more resilient to data staleness, as the increased data class overlap across clients

allows the fresh data of some clients to compensate for the outdated clients.

In Figure 11c, we observe that the payments in three χ are almost the same, because χ only controls the data heterogeneity and has no impact on the data collection cost.

APPENDIX K SYSTEM EFFICIENCY

We conduct our experiments on a Nvidia RTX 4090 GPU and quantify the actual system efficiency on each dataset in Table III.

Firstly, as shown in the first row of Table III, we record the *wall-clock runtime per time slot* (i.e., lines 2-16 in Algorithm 3) on three datasets and different temporal concept drift settings. We record the average consumed time across the whole process (i.e., 200 slots for MNIST [24] and Cifar-100 [25] datasets, 14 slots for CICIoMT [62] dataset). It shows that our mechanism can complete the client selection and pricing, and federated unlearning of each time slot in a short time.

Secondly, as shown in the second row of Table III, we record the *end-to-end FU time* (i.e., lines 1-17 in Algorithm 3) on three datasets, proving the efficiency of our mechanism.

Lastly, as shown in the last row of Table III, we record the *storage overhead from maintaining historical checkpoints*. In each dataset, the storage overhead from maintaining historical checkpoints is φ times the model size of the global model according to Algorithm 1, regardless of different temporal concept drift settings. The results in Table III show that the required storage overhead is acceptable even in edge devices with limited storage space.

APPENDIX L COMPARISON WITH EARLIER VERSION IN [41]

In this section, we use an example with three clients to demonstrate that (i) in the earlier version [41], all clients that will be selected with honest reporting will misreport for increasing their utility, (ii) none of the clients in this journal mechanism can obtain a larger utility by misreporting. Since the client selection and payment rule are the same for each time slot, we omit the time index t for simplicity. Further, since data criticality δ_i is the only private information in client's value θ_i and their relationship is linear in (21), we compare whether

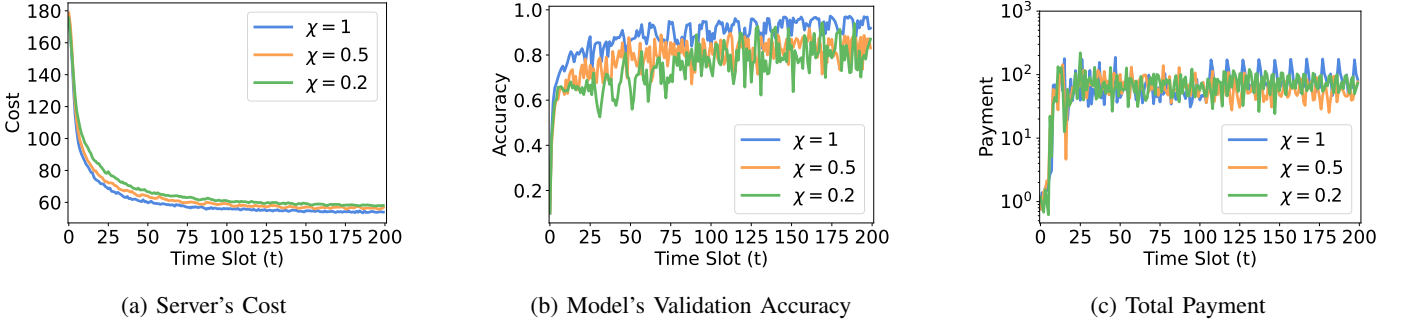
Figure 11: Performance of Various Non-IID Levels χ under OCSP mechanism

Table III: System Efficiency

Metric	MNIST [24]		Cifar-100 [25]		CICIoMT [62]
	Mislabeling	Noise	Mislabeling	Noise	Default
Average wall-clock runtime per time slot (s)	3.916	4.035	4.879	5.354	9.438
End-to-end FU time (s)	783.36	807.33	975.949	1070.849	132.161
Storage overhead from historical checkpoints (MB)	0.40		4.25		4.72

they honestly report their value θ_i or not for simplicity. We consider a set of clients $\mathcal{I} = \{1, 2, 3\}$ with true data collection costs $\{c_1, c_2, c_3\}$ and value metric $\{\theta_1, \theta_2, \theta_3\}$.

We first use Table IV to demonstrate that some clients can obtain larger utilities by misreporting in the earlier version [41]. In [41], the clients selection and payment rule consists of four steps: (i) each client i honestly reports $\beta_i = c_i - \theta_i$, (ii) the server computes the M -th smallest value $\beta[M]$ in $\beta \triangleq \{\beta_i\}_{i \in \mathcal{I}}$ and broadcasts it to all clients, (iii) each client i decides the price by

$$p_i = \begin{cases} = \theta_i + \min\{0, \beta_t[M]\}, & \text{if } \beta_i \leq 0 \text{ and } \beta_i \leq \beta_t[M], \\ \geq c_i, & \text{otherwise,} \end{cases} \quad (115)$$

and reports $g_i = \theta_i - p_i$ to the server, and (iv) the server selects clients by

$$a_i = \begin{cases} 1, & \text{if } g_i \geq 0 \text{ and } g_i \geq g_t(M), \\ 0, & \text{otherwise,} \end{cases} \quad (116)$$

where $g_t(M)$ is the M -th largest value of $g_t \triangleq \{g_{i,t}\}_{i \in \mathcal{I}}$.

(a) *Truthful Report in the Earlier Version [41]*. Following the earlier version [41], clients first submit a combined metric $\beta_i = c_i - \theta_i$ to the server. Without loss of generality, we assume $\beta_i \leq 0$ for all $i \in \mathcal{I}$, ordered such that $\beta_1 < \beta_2 < \beta_3$. Assuming a transmission capacity constraint of $M = 2$, the server identifies and broadcasts the payment threshold $\beta[M] = \beta_2$. Next, each client determines their asking price p_i using the derived pricing function and reports the second combined metric $g_i = \theta_i - p_i$ to the server. The server then selects the M clients with the largest g_i values (i.e., clients 1 and 2 in this case). The resulting utilities under truthful reporting are summarized in the final column under (a) *Truthful Report*.

(b) *Misreport in the Earlier Version [41]*. However, this truthful profile is highly unstable. Suppose clients 1 and 2

strategically misreport their private information by submitting inflated metrics $\beta'_1 > \beta_1$ and $\beta'_2 > \beta_2$, while still satisfying the condition $\beta'_1 < \beta'_2 < \beta_3$. Due to these misreports, the server broadcasts an inflated threshold $\beta'[M] = \beta'_2$. After calculating their new asking prices p'_i and reporting g'_i , clients 1 and 2 are still successfully selected. However, their utilities increase (as shown in the last column) because their compensation is now tied to the manipulated higher threshold β'_2 . Crucially, this example illustrates a general structural flaw: any selected client under truthful reporting can successfully misreport to extract a higher utility, provided they manipulate the new threshold $\beta'[M]$ such that $\beta[M] < \beta'[M] < \beta[M+1]$. Because a rational client can always exploit this gap to increase their utility while guaranteeing selection, truthful reporting is not a dominant strategy in the earlier version [41].

Next, we refer to Tables V, VI, and VII to demonstrate that clients cannot obtain a larger utility by misreporting in our journal mechanism. We consider the same true data collection cost profile $\{c_1, c_2, c_3\}$ and true value metric profile $\{\theta_1, \theta_2, \theta_3\}$ as in the earlier version [41]. In the journal mechanism, the server maintains an estimated value metric profile $\{\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3\}$ and a data size profile $\{d_1, d_2, d_3\}$. Without loss of generality, we assume the true metrics are ordered such that $\frac{\hat{\theta}_1 - c_1}{d_1} > \frac{\hat{\theta}_2 - c_2}{d_2} > \frac{\hat{\theta}_3 - c_3}{d_3}$.

Table V shows the baseline utilities under truthful reporting, where each client submits an honest bid $b_i = c_i$. Based on these bids, the server computes the selection metric $\hat{g}_i$ for each client. Under the transmission capacity constraint M , suppose the server selects clients 1 and 2, who possess the highest $\hat{g}_i$ values. The server then determines the payments p_i , yielding the resulting true utilities u_i detailed in the last column.

To show that truthful reporting is a dominant strategy, we analyze all possible deviations for both a winning client (client 1) in Table VI and a losing client (client 3) in Table VII. The

Table IV: Vulnerability Analysis of the earlier version [41]

Client	(a) Truthful Report				(b) Misreport			
	Report β_i	Price p_i	Report g_i	Utility u_i	Report β'_i	Price p'_i	Report g'_i	Utility u'_i
Client 1	$\beta_1 = c_1 - \theta_1$	$\theta_1 + \beta_2$	$-\beta_2$	$u_1 = \beta_2 - \beta_1 > 0$	β'_1	$\theta_1 + \beta'_2$	$-\beta'_2$	$\beta'_2 - \beta_1 > u_1$
Client 2	$\beta_2 = c_2 - \theta_2$	$\theta_2 + \beta_2$	$-\beta_2$	$u_2 = \beta_2 - \beta_2 = 0$	β'_2	$\theta_2 + \beta'_2$	$-\beta'_2$	$\beta'_2 - \beta_2 > u_2$
Client 3	$\beta_3 = c_3 - \theta_3$	$\geq c_3$	$\leq -\beta_3$	0	β_3	$\geq c_3$	$\leq -\beta_3$	0

Table V: Truthful Reporting Profile in this paper

Client	Estimated $\hat{\theta}_i$	Bid b_i	Compute $\hat{g}_i$	Payment p_i	True Utility u_i
Client 1	$\hat{\theta}_1$	c_1	$\frac{\hat{\theta}_1 - c_1}{d_1}$	$\hat{\theta}_1 - d_1 \hat{g}_3$	$\hat{\theta}_1 - d_1 \hat{g}_3 - c_1 > 0$
Client 2	$\hat{\theta}_2$	c_2	$\frac{\hat{\theta}_2 - c_2}{d_2}$	$\hat{\theta}_2 - d_2 \hat{g}_3$	$\hat{\theta}_2 - d_2 \hat{g}_3 - c_2 > 0$
Client 3	$\hat{\theta}_3$	c_3	$\frac{\hat{\theta}_3 - c_3}{d_3}$	0	0

deviations of client 1 can directly apply to client 2, so we omit client 2's analysis. The results of possible deviations are summarized into four cases:

(a) *Client 1 remains selected.* Suppose client 1 submits an untruthful bid $c'_1 \neq c_1$, while all other clients report c_2 and c_3 honestly. The server computes the selection metric $\hat{g}'_1$. If client 1 is still selected (i.e., its new metric remains above the threshold set by the rejected client 3 as $\hat{g}'_1 > \hat{g}_3$), its payment p'_1 remains exactly the same as in the truthful case. This is because the payment depends strictly on the critical value of the first rejected client 3, which client 1's bid cannot alter. Thus, client 1 cannot increase its utility.

(b) *Client 1 is not selected.* Suppose client 1 submits an inflated bid $c'_1 > c_1$ such that its selection metric $\hat{g}'_1$ drops below the critical threshold, resulting in rejection. Its payment p'_1 immediately becomes zero. Because its original utility u_1 was non-negative, this deviation strictly decreases its utility. Note that while the utilities of other clients may shift, under the unilateral deviation principle, we solely evaluate the misre-

porter's outcome.

(c) *Client 3 remains not selected.* Suppose client 3 submits an untruthful bid $c'_3 \neq c_3$ and other clients honestly report c_1 and c_2 . If client 3 remains unselected, its payment p'_3 and utility u'_3 remain zero. The deviation does not yield a larger utility to the misreporter.

(d) *Client 3 becomes selected.* Suppose client 3 underbids by $c'_3 < c_3$ to inflate its metric $\hat{g}'_3$ and win the auction. By forcing its way into the winning set, client 3 displaces another client, establishing a new and higher critical threshold $\hat{g}'_{thres} \in \{\hat{g}_1, \hat{g}_2\}$. The new utility for client 3 becomes $u'_3 = p'_3 - c_3 = \hat{\theta}_3 - d_3 \hat{g}'_{thres} - c_3$. However, because the new threshold is strictly greater than client 3's true metric ($\hat{g}'_{thres} > \hat{g}_3 = \frac{\hat{\theta}_3 - c_3}{d_3}$), substituting this yields $u'_3 < \hat{\theta}_3 - d_3 \hat{g}_3 - c_3 = \hat{\theta}_3 - d_3 \left(\frac{\hat{\theta}_3 - c_3}{d_3} \right) - c_3 = 0$. Thus, lowering the bid to win results in a strictly negative utility for client 3.

In summary, unlike the earlier version [41], no client can obtain a higher utility by misreporting their bid in this journal mechanism.

Table VI: Vulnerability Analysis of this paper (Client 1)

Client	(a) Client 1 remains selected			(b) Client 1 is not selected		
	Bid b'_i	g'_i	Payment p'_i	Bid b'_i	g'_i	Payment p'_i
Client 1	$c'_1 \neq c_1$	$\frac{\hat{\theta}_1 - c'_1}{d_1}$	$\hat{\theta}_1 - d_1 \hat{g}_3$	$c'_1 > c_1$	$\frac{\hat{\theta}_1 - c'_1}{d_1}$	0
Client 2	c_2	$\frac{\hat{\theta}_2 - c_2}{d_2}$	$\hat{\theta}_2 - d_2 \hat{g}_3$	c_2	$\frac{\hat{\theta}_2 - c_2}{d_2}$	—
Client 3	c_3	$\frac{\hat{\theta}_3 - c_3}{d_3}$	0	c_3	$\frac{\hat{\theta}_3 - c_3}{d_3}$	—

Table VII: Vulnerability Analysis of this paper (Client 3)

Client	(c) Client 3 remains not selected			(d) Client 3 becomes selected		
	Bid b'_i	g'_i	Payment p'_i	Bid b'_i	g'_i	Payment p'_i
Client 1	c_1	$\frac{\hat{\theta}_1 - c_1}{d_1}$	$\hat{\theta}_1 - d_1 \hat{g}'_3$	c_1	$\frac{\hat{\theta}_1 - c_1}{d_1}$	$\hat{\theta}_1 - d_1 \hat{g}'_{thres.} < p_1$
Client 2	c_2	$\frac{\hat{\theta}_2 - c_2}{d_2}$	$\hat{\theta}_2 - d_2 \hat{g}'_3$	c_2	$\frac{\hat{\theta}_2 - c_2}{d_2}$	$\hat{\theta}_2 - d_2 \hat{g}'_{thres.} < p_2$
Client 3	$c'_3 \neq c_3$	$\hat{g}'_3 = \frac{\hat{\theta}_3 - c'_3}{d_3}$	0	$c'_3 < c_3$	$\hat{g}'_3 = \frac{\hat{\theta}_3 - c'_3}{d_3}$	$\hat{\theta}_3 - d_3 \hat{g}'_{thres.} < c_3$